this excited state does not last longer than 10^{-12}s and decay into two daughter nuclei along with the release of 2 or 3 neutrons. In each reaction, on an average, 2.5 neutrons are emitted. It is shown in Figure 9.27

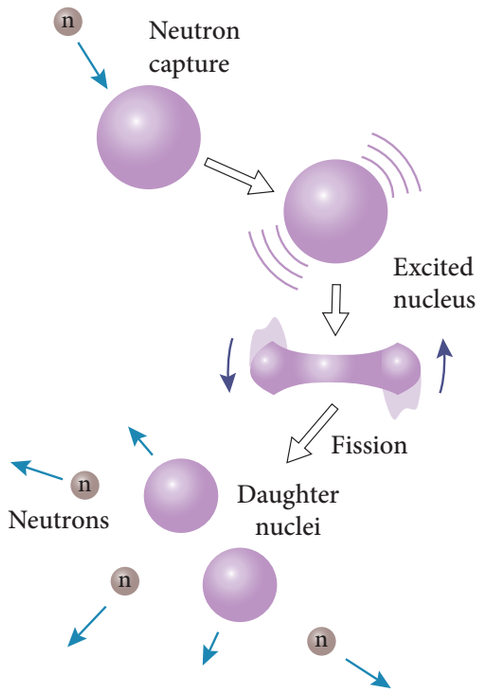


Figure 9.27 Nuclear fission

Energy released in fission:

We can calculate the energy (Q) released in each uranium fission reaction. We choose the most observed fission reaction which is given in the equation (9.45).

$$^{235}_{92}U + {}^{1}_{0}n \rightarrow {}^{236}_{92}U^{*} \rightarrow {}^{141}_{56}Ba + {}^{92}_{36}Kr + 3\,{}^{1}_{0}n + Q$$

Mass of $^{235}_{92}U$	= 235.045733 u
Mass of $^{1}_{0}n$	= 1.008665 u
Total mass of reactants	= 236.054398 u
Mass of $^{141}_{56}Ba$	= 140.9177 u
Mass of $^{92}_{36}Kr$	= 91.8854 u
Mass of 3 neutrons	= 3.025995 u
The total mass of products	= 235.829095 u

Mass defect $\Delta m = 236.054398\,u - 235.829095\,u$
$= 0.225303\,u$

So the energy released in each fission = $0.225303 \times 931 MeV \approx 200\,MeV$

This energy first appears as kinetic energy of daughter nuclei and neutrons. But later, this kinetic energy appears in the form of heat given to the surrounding.

Chain reaction:

When one $^{235}_{92}U$ nucleus undergoes fission, the energy released might be small. But from each fission reaction, three neutrons are released. These three neutrons can cause further fission in three other $^{235}_{92}U$ nuclei which in turn produce nine neutrons. These nine neutrons initiate fission in another 27 $^{235}_{92}U$ nuclei and so on. This process is called a chain reaction and the number of neutrons goes on increasing almost in geometric progression. It is shown in Figure 9.28.

There are two kinds of chain reactions: (i) uncontrolled chain reaction (ii) controlled chain reaction. In an uncontrolled chain reaction, the number of neutrons multiply indefinitely and the entire amount of energy is released in a fraction of second.

The atom bomb is an example of nuclear fission reaction in which uncontrolled chain reaction occurs. Atom bombs produce massive destruction on mankind. During World War II, on August 6 and 9 in the year 1946, USA dropped two atom bombs in two places of Japan, Hiroshima and Nagasaki. As a result, lakhs of people were killed and the two cities were completely destroyed. Even now the people who are living in those places have side effects caused by the explosion of atom bombs.

It is possible to calculate the typical energy released in a chain reaction. In the first step,

UNIT-9(XII-Physics_Vol-2)_New.indd 180 21-07-2020 15:50:40